

Anthony Coache

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Degrees

- Ph.D.**, Statistical Sciences (GPA: 3.95/4.00), University of Toronto Oct 2024
• *Thesis: Risk-Sensitive Reinforcement Learning With Dynamic Risk Measures*
- M.Sc.**, Statistics (GPA: 4.30/4.30), Université du Québec à Montréal Jun 2020
• *Thesis: Stochastic Portfolio Optimization under Coherent Risk Measures*
- B.Sc.**, Mathematics & Statistics (GPA: 4.18/4.30), Université du Québec à Montréal Sep 2017

Work Experience

- Imperial College London, Research Associate** Jun 2024 – Jun 2026
• Led and collaborated on 4 concurrent projects in mathematical finance spanning risk modelling, statistical learning, optimisation, and market microstructure
• Supervised M.Sc. research projects conducted with leading banks and hedge funds, guiding model development and quantitative analysis
• Designed and taught courses in quantitative risk management and optimisation
- Oxford-Man Institute, University of Oxford, Visiting Researcher** Jan 2022 – Jun 2022
• Developed a computationally efficient algorithm for risk-sensitive RL, leveraging elicibility to avoid nested simulations
• Invitation for a 6-month research visit from Prof. Álvaro Cartea
- Collaboration between Oanda and Fields-CQAM, Research Assistant** Jan 2020 – Sep 2020
• Analyzed trends in order-flow trading data using hidden Markov models
• Backtested dynamic hedging strategies, improving Sharpe and tail-VaR
- University of Toronto, Ph.D. Candidate** Sep 2019 – Jul 2024
• Published novel results in risk management and reinforcement learning with applications to dynamic decision-making under uncertainty
• Teaching assistant for 6 courses in statistics and financial insurance
- Université du Québec à Montréal, Research Intern** May 2016 – Aug 2017
• Interdisciplinary works in statistics, portfolio optimization and epidemiology
• NSERC Undergraduate Student Research Awards & FRQNT Supplements

Publications

- A. Capponi, **A. Coache**, J. Muhle-Karbe. (2026) Optimal Trading in Automated Market Makers. **Submitted to INFORMS**.
- M. Moghimi, **A. Coache**, H. Ku. (2026) Decoupling Time and Risk: Risk-Sensitive Reinforcement Learning with General Discounting. **Submitted to NeurIPS**.
- Z. Cheng, **A. Coache**, S. Jaimungal. (2025) Eliciting Risk Aversion with Inverse Reinforcement Learning via Interactive Questioning. **Revised and resubmitted at JMLR**.
- A. Coache**, S. Jaimungal. (2025) Robust Reinforcement Learning with Dynamic Distortion Risk Measures. **SIAM J. Math. of Data Sci.**
- A. Coache**, S. Jaimungal, Á. Cartea. (2024) Conditionally Elicitable Dynamic Risk Measures for Deep Reinforcement Learning. **SIAM J. Financial Mathematics**.
- A. Coache**, S. Jaimungal. (2023) Reinforcement Learning with Dynamic Convex Risk Measures. **Mathematical Finance**.

Technical Skills

- *Methods: RL, risk measures, stochastic modeling, optimization, DeFi, applied statistics, piano*
- *Programming: Python (PyTorch, Pandas, NumPy, matplotlib, sklearn, SciPy), TeX, Matlab, R, SQL*

Selected Scholarships & Awards

Mathematical Finance (Wiley) Top Cited Article of 2023 Publications	Mar 2025
NSERC Postdoctoral Fellowship	Sep 2024 – Aug 2026
UofT DoSS Doctoral Early Research Excellence Award	Jun 2023
SIAG/FME Conference Paper Prize	Jun 2023
Oxford-Man Institute Visitors Programme	Jan 2022
NSERC Alexander Graham Bell + FRQNT + OGS Doctoral Awards	Sep 2019 – Aug 2024
NSERC Alexander Graham Bell + FRQNT Master's Awards	Sep 2017 – Aug 2019

Selected Invited Talks

<i>London-Paris Bachelier Workshop</i> , London, United Kingdom	Nov 2025
<i>SIAM Conference on Financial Mathematics and Engineering</i> , Miami, USA	Jul 2025
<i>ETH-HKG-ICL Mathematical Finance Workshop</i> , Hong Kong, China	Apr 2025
<i>King's College London Mathematical Finance Seminar</i> , London, United Kingdom	Feb 2025
<i>Control and Optimization Seminar at University of Connecticut</i> , Online	Nov 2024
<i>Mathematical Finance Seminar at Illinois Institute of Technology</i> , Online	Oct 2024
<i>Mathematical Insights from Markets, Control and Learning</i> , Aussois, France	Sep 2024
<i>STATQAM Seminar</i> , Montréal, Canada	Feb 2024
<i>SIAM Conference on Financial Mathematics and Engineering</i> , Philadelphia, USA	Jun 2023
<i>INFORMS Annual Meeting</i> , Indianapolis, USA	Oct 2022
<i>SIAM Conference on Financial Mathematics and Engineering</i> , Online	Jun 2021

Teaching Experience

Imperial College London

- *Lecturer: Optimisation in Machine Learning* Autumn 2025
- *Lecturer: Quantitative Risk Management* Autumn 2024 & 2025
- *Lecturer: Convex Optimisation* Autumn 2024

University of Toronto

- *Course instructor: Stochastic Methods for Actuarial Science* Fall 2023
- *Workshop leader: Python and Matlab bootcamps for MFI program* Fall 2021 & 2022
- *Teaching assistant: 6 courses in statistics and financial insurance* Sep 2020 – Apr 2024

Université du Québec à Montréal

- *Teaching assistant: 4 courses in statistics* Sep 2016 – Dec 2019

Student Supervision

M.Sc. Thesis at Imperial College London

- *Matthieu Fisher (Eisler Capital), Antoine Fuentes (BNP Paribas), Leo Goedhart, Zheyuan Li, Akram Zakriti (ExodusPoint)* Summer 2025
- *Jeanne Gauthier (Morgan Stanley), Eloi Godier (Sakana Research), Linze Li (Ocean Leonid), Oussama Saadi (Deutsche Bank), Dean Yang (Qube)* Summer 2024

Academic Service

Journal Referee

- *Quant. Finance* (x3), *SIAM J. Control and Optim.* (x1), *IMA J. Math. Control and Information* (x1)
- *NeurIPS Workshop Generative AI in Finance* (x5), *International Conference on AI in Finance* (x17)

Conference Organizing Committee

- *Finance & Stochastics Seminar 2025-2026 at Imperial College London (organiser)*
- *Canadian Statistics Student Conference 2020 & 2021 (translation, session chair)*
- *First Edition of the Statistics Student Summit in Montréal 2019 (co-chair)*

Last updated on June 22, 2026. Full academic CV available upon request.